VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	

Continuous Assessment Test (CAT) – I Jan 2026

Programme	:	B.Tech (BAI,BCE,BRS,BPS)	Semester	:	IV
Course Code & Course Title	:	BCSE204L & Design and Analysis of Algorithms	Class Number	:	CH2025260501137 CH2025260501139 CH2025260501147 CH2025260501150 CH2025260501152 CH2025260501153
Faculty	:	Dr. Kaja Mohideen A Dr. Manimegalai Dr. Pandiyaraju V Dr. Senthil Kumar A M Dr. Raja Sree T Dr. Suguna M	Slot	:	A2+TA2
Duration	:	1 hour 30 Minutes	Max. Mark		50

Answer all questions

Q. No	Sub Sec.	Description	Marks
1		You are given an array $A[1..n]$ of n positive integers. The objective is to determine the maximum absolute difference between any two distinct elements $A[i]$ and $A[j]$, where $i < j$. a) Write a brute-force algorithm to compute the maximum absolute difference by checking all valid pairs of elements in the array. [4 Marks] b) For your algorithm, express its time complexity using Big-O, Big-Ω, and Big-Θ notations. [2 Marks] c) Design an optimized algorithm using the Insertion Sort concept: first sort the array using Insertion Sort, then compute the maximum absolute difference as $A[n] - A[1]$ from the sorted array. State the overall time complexity and briefly justify why the brute-force approach is inefficient for large inputs. [4 Marks]	10
2		Perform the following tasks: Given a sequence of n integers $S = [s1, s2, \dots, sn] = [2, -3, 4, -1, -2, 5, -3, 2, 1]$ the task is to compute the Maximum Subarray Product (MSP) defined as $max_{1 \le i \le j \le n} \prod_{k=i}^{j} S_k$ where $si, si+1, \dots, sj$ is a contiguous subarray of S. If there exist two or more subarrays producing the same Maximum Product (MP), select the subarray with the minimum length $((j - i + 1)$ as the final result. Perform the following tasks: a) Design a Divide and Conquer algorithm to compute the MSP, using the minimum-length subarray as a tie-breaker when products are equal. [6 Marks] b) Illustrate the algorithm on the given sequence and show the intermediate steps, including the candidate subarrays considered and their products. [5 Marks]	15

		c) Analyze the time complexity of the algorithm and state the final MSP, the selected subarray, and its length. [4 Marks]	
3		An airline cargo service operates a small aircraft with a maximum load capacity of 50 kg. The airline receives multiple cargo items, each with a specific weight and profit. To maximize profit, the airline is allowed to load items fractionally if the full item cannot be accommodated. The objective is to select items or fractions of items to maximize the total profit without exceeding the weight limit. The details of the available cargo items are given below: Item | Weight (kg) | Profit (₹) A | 10 | 6 B | 20 | 10 C | 30 | 12 D | 15 | 9 E | 25 | 13 F | 5 | 4 a) Determine which cargo items or fractions of items should be loaded into the aircraft to maximize the total profit without exceeding 50 kg. Show all necessary steps and calculations clearly. [5 Marks] b) Write the pseudocode for the Fractional knapsack algorithm and state its time complexity. [5 Marks]	10
4		Given five matrices A_1, A_2, A_3, A_4, A_5 with dimensions 5×10, 10×3, 3×12, 12×5, and 5×50 respectively, determine the minimum number of scalar multiplications required to multiply the entire chain $A_1A_2A_3A_4A_5$ using the dynamic programming approach. Construct the dynamic programming cost table $m[i,j]$ and the split table $k[i,j]$, showing the minimum multiplication cost and the corresponding split point for every subchain. Finally, identify the optimal parenthesization that achieves the minimum cost and analyse the time complexity. [Logic: 2 marks, Pseudocode: 4 marks, Illustration: 6 marks, Time complexity analysis: 3 marks.]	15

**********All the best **********